

Education

Nanjing University Sep 2023 – present
B.Eng. in Intelligent Software Engineering (expected Jun 2027) | **Average:** 88.4/100
School of Intelligent Software and Engineering

Research Interests

AI systems that continually improve from data, experience, and human feedback.
Data-centric AI • Self-improving agents and continual learning • Human–AI interaction

Research Experience

The Hong Kong University of Science and Technology (Guangzhou) Aug – Oct 2026
Visiting Research Student | Research with Prof. [Weikai Yang](#)

Nanjing University, iMATE Lab Mar 2025 – present
Research Assistant | Advisor: Prof. Ke Xu; collaborator: Dr. [Wenchao Li](#)
School of Intelligence Science and Technology

Research and Publications

SliceTune: Supporting Interactive Intent-Guided Slice Finding for Model Validation
First author; research lead | CHI 2027 submission, under review

- Designed and built **intent-guided failure-slice discovery**, mapping validation goals to data attributes through a Feature Graph. Led evaluation with image, audio, and text case studies, a controlled user study, and expert interviews to assess failure discovery, diagnosis, and data-guided model improvement.

SkillContour: Steering Agent Skill Consolidation with Behavioral Evidence
First author; research lead | CHI 2027 submission, under review

- Built an **evidence-linked skill refinement workbench** connecting source patches, skill instructions, and execution traces to support regrouping, re-synthesis, and replay. Led user and task evaluations showing stronger diagnostic support and higher agent accuracy than automatic consolidation and text-based review.

SkillCommit: Evolving Agent Skills through Behaviorally Validated Scope Expansion
Yu He, Weikai Yang.
First author; research lead | ARR submission, under review | [arXiv 2026](#)

- Designed **replay-validated skill consolidation** using cross-instance replay and mechanism checks to assess compatibility before committing merged skills. Demonstrated gains in rule-based, legal, and abstract reasoning without parameter updates, with learned skills transferring across model sizes and families.

NetworkCanvas: Supporting Progressive Network Visualization Exploration via Adaptive Recommendations
Wenchao Li, Yuewen Gao, **Yu He**, Cong Zhu, Ke Xu.
[CHI 2026](#) | Published | Project: Mar – Sep 2025

- Implemented **visualization and feedback components** for adaptive network exploration and branching histories; contributed to data integration, figures, and study documentation.

MSSR: Memory-Aware Adaptive Replay for Continual LLM Fine-Tuning
Yiyang Lu, **Yu He**, Jianlong Chen, Hongyuan Zha.
[arXiv preprint, 2026](#) | Project: Aug 2025 – Jan 2026

- Modeled **sample-level knowledge retention** and implemented adaptive replay scheduling; evaluated continual fine-tuning, with gains over replay baselines, especially in reasoning.

Selected Honors and Awards

Nanjing University People’s Scholarship Nov 2025
Meritorious Winner, Mathematical Contest in Modeling (COMAP) May 2025
National Third Prize, China Collegiate Innovation and Invention Competition Dec 2024